

NIMBY Risk Research

****A curated evidence base for community opposition risk in infrastructure siting.****

This repository documents the peer-reviewed academic research underpinning LANDMARQ's community opposition probability model. It maps published findings from energy policy, urban planning, and environmental justice to the seven variable categories that drive the NIMBY scoring engine.

Built and maintained by [Beatriz Mendoza](https://github.com/beatriz-mendoza), Co-Founder & CEO of [LANDMARQ](https://landmarq.io).

Why this exists

Infrastructure developers lose an estimated ****\$156B annually**** to community opposition – projects killed at town halls, delayed by injunctions, or abandoned after \$200K–\$800K of diligence capital has already been committed. The pattern is consistent, costly, and almost entirely predictable from publicly available data.

The academic literature on infrastructure siting, community opposition, and permitting outcomes has documented the key predictors for over two decades. This repository makes that evidence base accessible in one place – organized by variable category, summarized for practitioners, and linked to the original sources.

This is not a black box. Every variable in LANDMARQ's model traces back to published research. This repository is the audit trail.

Repository structure

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nimby-risk-research/
├── README.md
├── evidence-base/
│   ├── 01-community-sentiment.md      # Variable 1: strongest
├── predictor
│   ├── 02-historical-permitting.md    # Variable 2: Bayesian prior
│   ├── 03-regulatory-zoning.md       # Variable 3: institutional
├── bottleneck
│   ├── 04-community-equity.md         # Variable 4: benefit-sharing
├── signal
│   ├── 05-physical-site.md           # Variable 5: proximity &
└── visibility
```

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|   |— 06-cumulative-burden.md           # Variable 6: saturation
effect
|   |— 07-demographics.md               # Variable 7: secondary signal
|— literature-map.md                     # Full citation index
|— methodology-notes.md                 # How research maps to model
weights
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The 7 variable categories

#	Variable	Predictive Weight	Key Research
1	Community Sentiment & Political Climate	Highest	Stokes et al. (2023, PNAS)
2	Historical Permitting Outcomes	High	Susskind et al. (2022, Energy Policy)
3	Regulatory & Zoning Environment	High	Wolsink (2000); Devine-Wright (2005)
4	Community Equity & Engagement	Medium	Löw-Beer et al. (2025, Nature Comms)
5	Physical & Site Characteristics	Medium	Roddis et al. (2018, Applied Energy)
6	Cumulative Burden	Secondary	US EPA (2022)
7	Demographic & Socioeconomic Composition	Secondary	Stokes et al. (2023, PNAS)

Weights reflect relative predictive contribution to opposition probability – not a fixed formula, but a Bayesian weighting calibrated against 30 completed US utility-scale projects.

How to use this

****Developers and investors**** – use this to understand what drives a LANDMARQ score. Every variable category has a documented research basis. If a site scores Tier 3, this repo explains which variables are driving that classification and why the academic literature supports that weighting.

****Researchers**** – this is an applied mapping of academic findings to a commercial diligence context. Gaps and methodological limitations are documented in each file.

****Competitors**** – the model's defensibility comes from the labeled outcome dataset and calibration, not the variable selection. Variable selection is public knowledge. The data is not.

Related

- [LANDMARQ Platform](https://landmarq.io/platform-overview)
- [Methodology Disclaimer](https://landmarq.io/disclaimer)
- [landmarq-methodology](https://github.com/beatriz-mendoza/landmarq-methodology) – CRC formula and scoring logic
- [infrastructure-siting-data](https://github.com/beatriz-mendoza/infrastructure-siting-data) – public data analysis

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